

Parent Functions II: x^3 and $\sqrt[3]{x}$

TEACHER KEY | Algebra II Honors | Unit 1 | Day 5 | TEK A2.2.A

Objective: I can graph and analyze the cubic and cube root parent functions, prove algebraically that each is an odd function, and describe how their rates of change differ.

Vocabulary

Odd function — satisfies $f(-x) = -f(x)$ for every x . Its graph has **180° rotational** symmetry about the origin.

Even function — satisfies $f(-x) = f(x)$. Its graph is symmetric across the **y-axis**.

Average rate of change on $[a, b] = (f(b) - f(a)) / (b - a)$

Part 1 — Two Tables, One Relationship

x	-8	-2	-1	0	1	2	8
f(x) = x^3	-512	-8	-1	0	1	8	512

x	-8	-2	-1	0	1	2	8
g(x) = $\sqrt[3]{x}$	-2	-1.26	-1	0	1	1.26	2

Find a pair of entries showing the two functions undo each other: **$f(2) = 8$ and $g(8) = 2$**

Part 2 — Prove Both Are Odd

Substitute $-x$ into each rule and simplify. Do not just check one number.

a. $f(x) = x^3$. Find $f(-x)$:

$f(-x) = (-x)^3 = -x^3 = -f(x)$ so f is **odd**

b. $g(x) = \sqrt[3]{x}$. Find $g(-x)$:

$g(-x) = \sqrt[3]{(-x)} = -\sqrt[3]{x} = -g(x)$ so g is **odd**

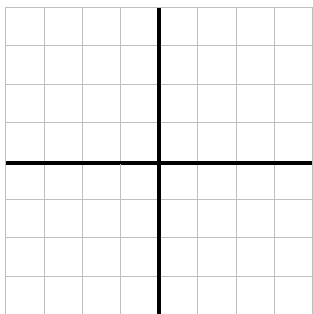
c. Test $h(x) = x^2$ the same way. Is it odd, even, or neither? **even, since $(-x)^2 = x^2$**

d. What does "odd" tell you about the graph without graphing it?

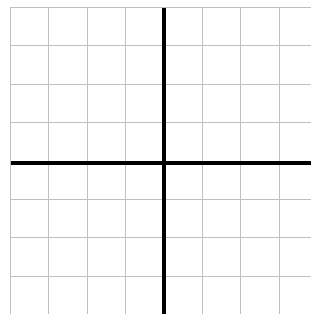
Part 3 — Graph Them Both

Both grids run -8 to 8 on each axis, so 1 square = 2 units.

A. $f(x) = x^3$



B. $g(x) = \sqrt[3]{x}$



Domain **all reals** Range **all reals**

Mirror line between them: **$y = x$**

Part 4 — Rates of Change

Compute the average rate of change of $f(x) = x^3$ on each interval.

On $[0, 1]$: **1** On $[1, 2]$: **7** On $[2, 3]$: **19**

Now $g(x) = \sqrt[3]{x}$ on $[0, 1]$: **1** On $[1, 8]$: **$1/7 \approx 0.14$**

Describe the contrast in one sentence:

The cubic **steepens** as x grows while the cube root **flattens** — which is what makes them mirror images.

Part 5 — Going Further

1. Solve $x^3 = 125$ and $x^2 = 25$. Count the real solutions for each.

$x^3 = 125 \rightarrow$ **$x = 5$, one solution** $x^2 = 25 \rightarrow$ **$x = \pm 5$, two solutions**

2. Explain the difference using odd and even.

x^3 is odd and never repeats an output; x^2 is even, so **x and $-x$ share an output**

3. Which of the four parent functions so far accept negative inputs? **$|x|$, x^3 , $\sqrt[3]{x}$**

4. Can a function be both even and odd? Find one or prove none exists.

Yes — **$f(x) = 0$, the only one**

Exit Ticket

A function has domain and range all reals, passes through the origin, and satisfies $f(-x) = -f(x)$.

Name two parent functions this could be: **x^3 and $\sqrt[3]{x}$**

Prove your choice is odd, algebraically: **$f(-x) = (-x)^3 = -x^3 = -f(x)$**